Instant Audio Tools

Documentation

v1.0

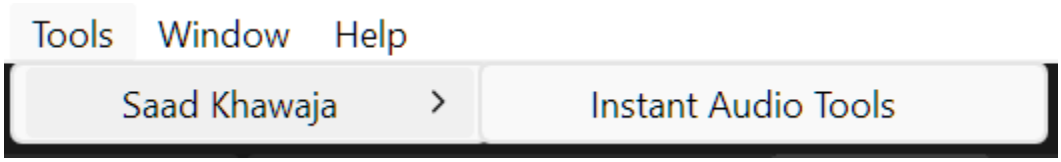
🎧 Instant Audio Tools — User Guide

Instant Audio Tools is a fast, non-destructive audio editing utility for Unity that lets you visually trim, fade, normalize, and preview audio clips directly inside the Editor — without leaving Unity or using external software.

Designed for **game developers, sound designers, and technical artists**, it provides an intuitive waveform editor with real-time preview and precise control.

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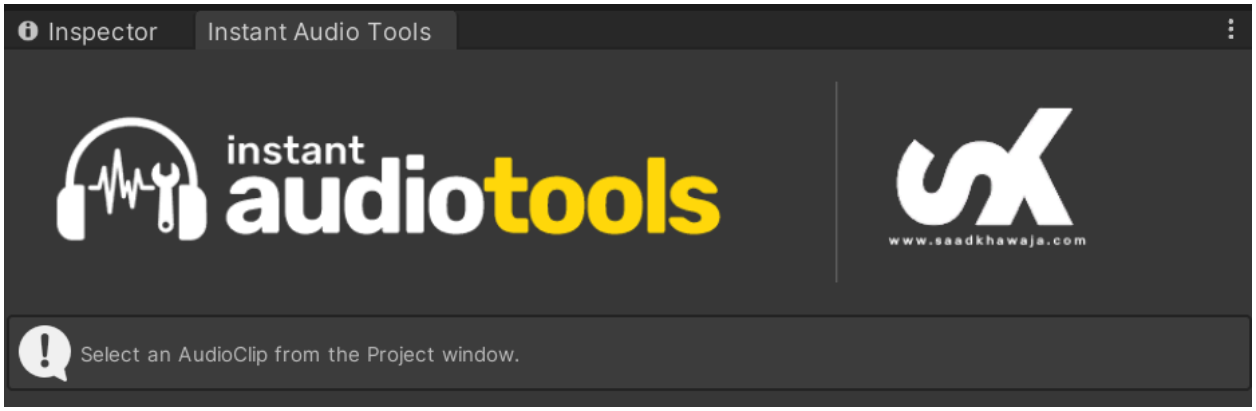
How to open the tools window



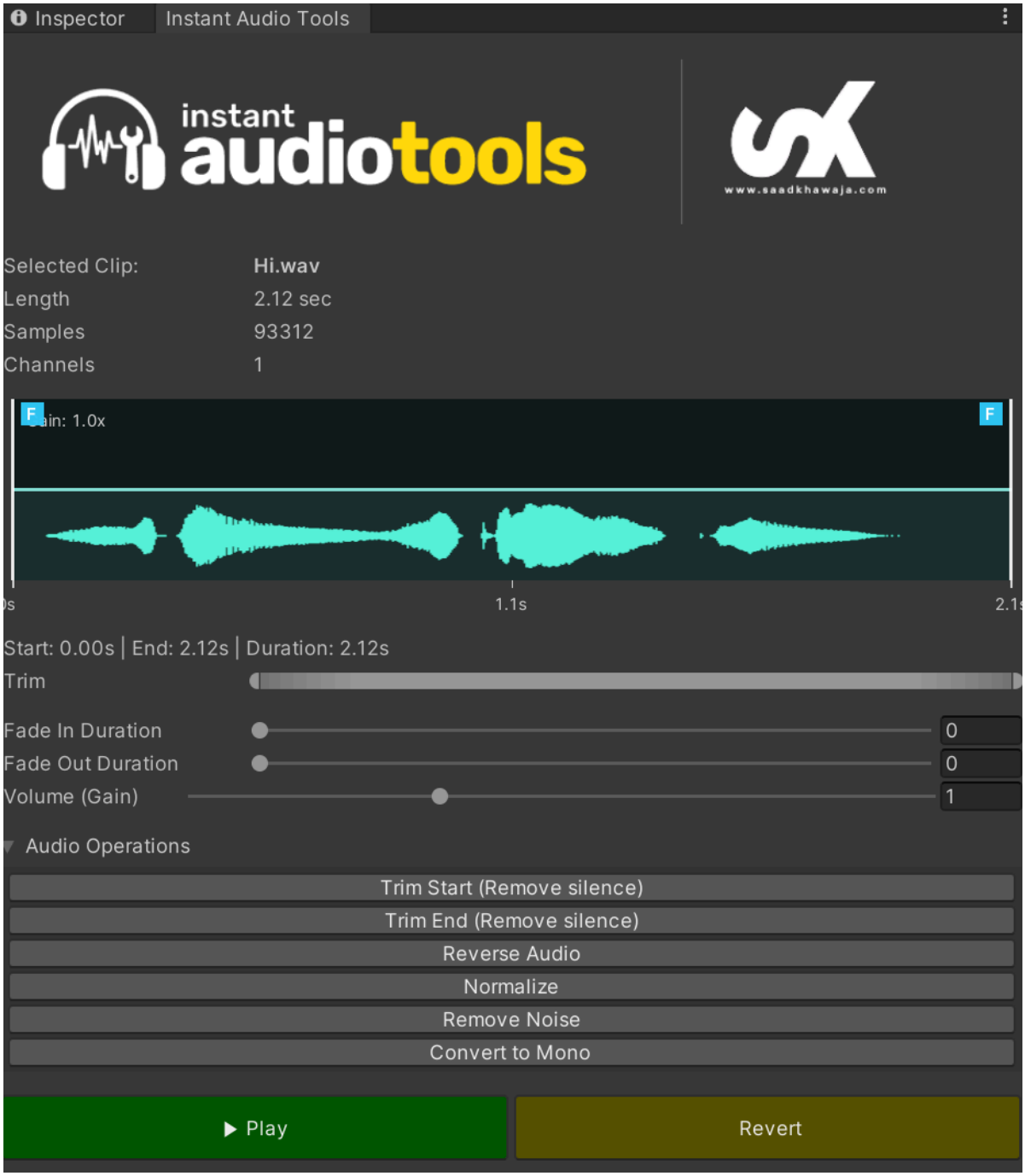
Tools -> Saad Khawaja -> Instant Audio Tools

To open the **InstantAudioTools** window, start by navigating to the main menu at the top of your screen. From there, select Tools, then move your cursor down to the **Saad Khawaja** option. Finally, click on Instant Audio Tools from the submenu that appears. This will open the Tools window for you to access various audio-related functions.

You will then see the editor window that says:



You need to select an audio file from the project window to see the editing options now.


Inspector
Instant Audio Tools
instant audiotools
www.saadkhawaja.com
Selected Clip: Hi.wav
Length 2.12 sec
Samples 93312
Channels 1
Gain: 1.0x
1.1s
Start: 0.00s | End: 2.12s | Duration: 2.12s
Trim
Fade In Duration 0
Fade Out Duration 0
Volume (Gain) 1
Audio Operations
Trim Start (Remove silence)
Trim End (Remove silence)
Reverse Audio
Normalize
Remove Noise
Convert to Mono
Play
Revert

🖼️ Interface Overview

The window is divided into five main sections:

1. **Header & Clip Info**
2. **Waveform Editor**
3. **Trim, Fade & Volume Controls**
4. **Audio Operations**
5. **Playback & Export**

ℹ️ Clip Information

At the top, you'll see:

- **Selected Clip name**
- **Length (seconds)**
- **Samples**
- **Channels**

This updates automatically when you select a different AudioClip.

🌊 Waveform Editor (Core Feature)

The waveform preview is fully interactive and shows the entire clip timeline.

🔊 Volume (Gain Line)

- The **horizontal line** represents volume (gain)
- Default **1.0x** = waveform fits into **half** the vertical space
- Drag the line:
    - Up → increase gain (up to 3.0x)
    - Down → reduce gain (to 0)

This does **not** permanently modify the clip unless saved.

✂ Trim (Start & End)

- **Vertical white lines** represent start and end trim points
- Drag:
    - Left line → trim start
    - Right line → trim end
- Darkened areas show **excluded audio**
- Trim is **non-destructive** until saved

You can also trim using the **Min-Max slider** below the waveform.

🎚 Fade In / Fade Out

Fade is handled visually and intuitively:

- **Blue triangular overlays** indicate fade areas
- **F handles** (small boxes with “F”) control fades

Fade In

- Left “F” handle
- Drag right → increase fade-in duration

Fade Out

- Right “F” handle
- Drag left → increase fade-out duration

Fade handles use **adaptive drag sensitivity**
(larger fades adjust faster when reducing)

⚠️ Fade **does not distort the waveform** visually — it uses highlight overlays for clarity.

🎛️ Controls Section

Trim

Shows:

`Start | End | Duration`

Includes a slider to fine-tune selection numerically.

Fade In / Fade Out Duration

Sliders allow precise numeric control in seconds.

Fade values are automatically clamped so:

- Fade In + Fade Out never exceed trimmed length

Volume (Gain)

- Slider + numeric field
- Range: `0.0 → 3.0`
- Affects preview and export

⚙ Audio Operations

Expandable **Audio Operations** panel includes:

- **Trim Start (Remove Silence)**
- **Trim End (Remove Silence)**
- **Reverse Audio**
- **Normalize**
- **Remove Noise**
- **Convert to Mono**

These operations:

- Modify the audio buffer
- Are tracked internally
- Can be reverted before saving

▶ Playback Controls

Play

- Previews only the **trimmed + faded** section
- Applies volume and fades in real time
- Does **not** overwrite the original clip

Revert

- Resets all changes:
    - Trim
    - Fade
    - Volume
    - Operations
- Restores original clip data

💾 Saving & Exporting

Save and Overwrite (WAV only)

- Replaces the original `.wav` file
- Disabled for non-WAV formats
- Automatically refreshes the asset

Save Audio As

- Saves a new `.wav` file
- Keeps original clip untouched
- Ideal for iterative sound design

🧠 Best Practices

- Use **WAV files** for overwrite support
- Use **Save Audio As** for safety

- Apply destructive operations last
- Preview before saving

⚠ Notes & Limitations

- MP3/OGG cannot be overwritten (Unity limitation)
- Tool is **Editor-only**
- Designed for **Unity 2021+**
- Fade & volume previews are accurate but lightweight for performance

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🧑‍💻 About & Support

Instant Audio Tools
Created by **Saad Khawaja**
🔗 https://www.saadkhawaja.com

For questions and help:

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